

HealthBridge Architecture

An ASHA-operated, ABDM/FHIR-linked platform that follows a rural patient across every level of the public health system â€” sub-centre, PHC, district hospital â€” with explainable danger-sign triage that reorders the queue and escalates emergencies.

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01The governing decision: real vs. mocked

Eight modules can't all be built for real in hackathon time â€” and shouldn't be. One real vertical spine, demoed end to end, beats eight shallow ones. Every module mocked below sits behind an adapter interface, so the mock is a swap-in later, not a rewrite.

●BUILD REAL

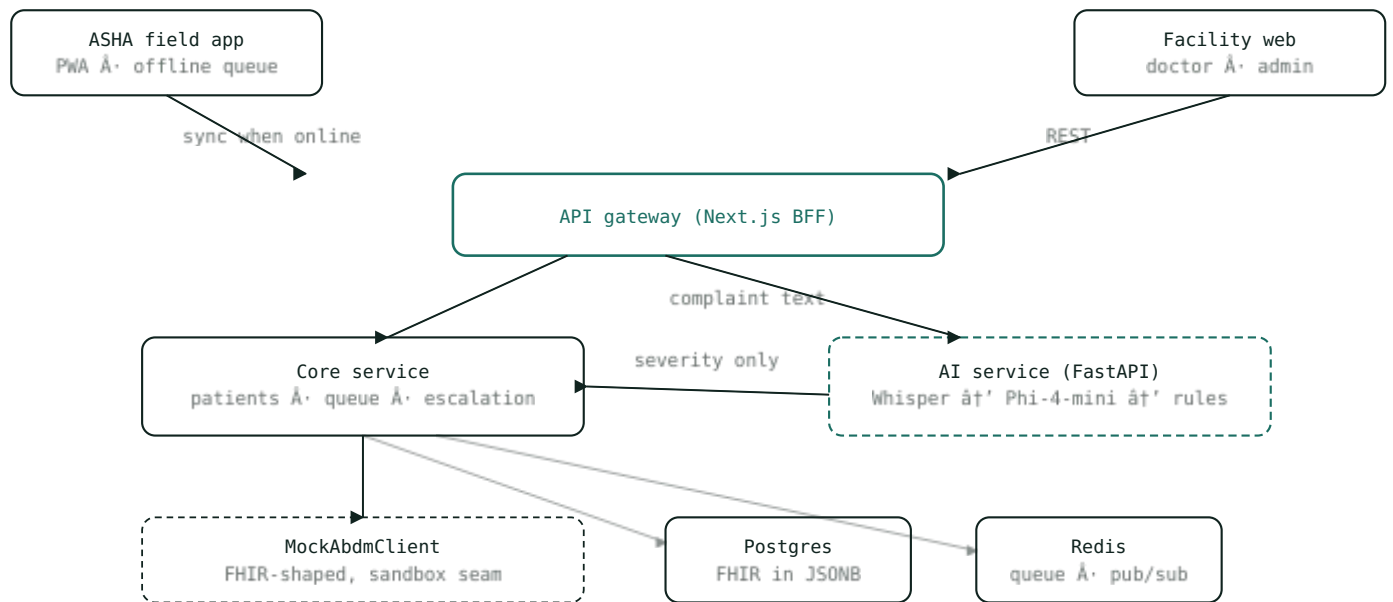
- **Patient record** â€” FHIR-shaped, keyed by ABHA number
- **Digital triage engine** â€” LLM extraction + IMNCI rule table
- **Severity-based queue** + red-case auto-escalation
- **Facility dashboard** â€” 4 tiles, real system data only
- **Offline-first ASHA app** â€” PWA with local sync

●CONVINCING MOCK, BEHIND AN ADAPTER

- **Live ABDM gateway** â†’ `MockAbdmClient` returns sandbox-shaped FHIR bundles
- **Teleconsult video** â†’ store-and-forward only, no live infra
- **Scheme verification** â†’ a `scheme_status` badge, not an engine
- **Referral, diagnostics, medicine stock, follow-up** â†’ data screens only

02System architecture

Two client surfaces, one backend, split into a deterministic Core service and a separate AI service â€” so a slow or offline model can never block a patient record from saving.



Both clients talk only to the gateway. The gateway splits deterministic work (Core, always available) from AI work (dashed border, degrades independently). The AI service returns a severity label “ never writes the record itself.

03Confirmed stack

Qdrant and GPT-5.6 were cut “ no vector-search use case for a deterministic rule engine, and a cloud LLM breaks offline-first and raises health-data compliance questions an SIH judge will ask.

FRONTEND

Next.js (App Router)

TypeScript

Tailwind CSS

One framework for both app surfaces and the BFF.

BACKEND

FastAPI

Python

REST APIs

Typed request/response models map cleanly onto structured triage output.

DATA

PostgreSQL

Redis

Qdrant “ cut

Postgres holds FHIR JSONB; Redis runs the priority queue and escalation pub/sub. No semantic search in the critical path.

Phi-4-mini via Ollama

LangGraph

Whisper (local)

GPT-5.6 “cut from critical path

Runs on modest PHC hardware with zero connectivity. Cloud LLM stays optional, non-critical, online-only.

04Data model

Clinical data is stored as FHIR resources so "ABDM/FHIR interoperable" is demonstrable, not a slide claim. Everything hangs off one key: the ABHA number.

FHIR resources (JSONB in Postgres)

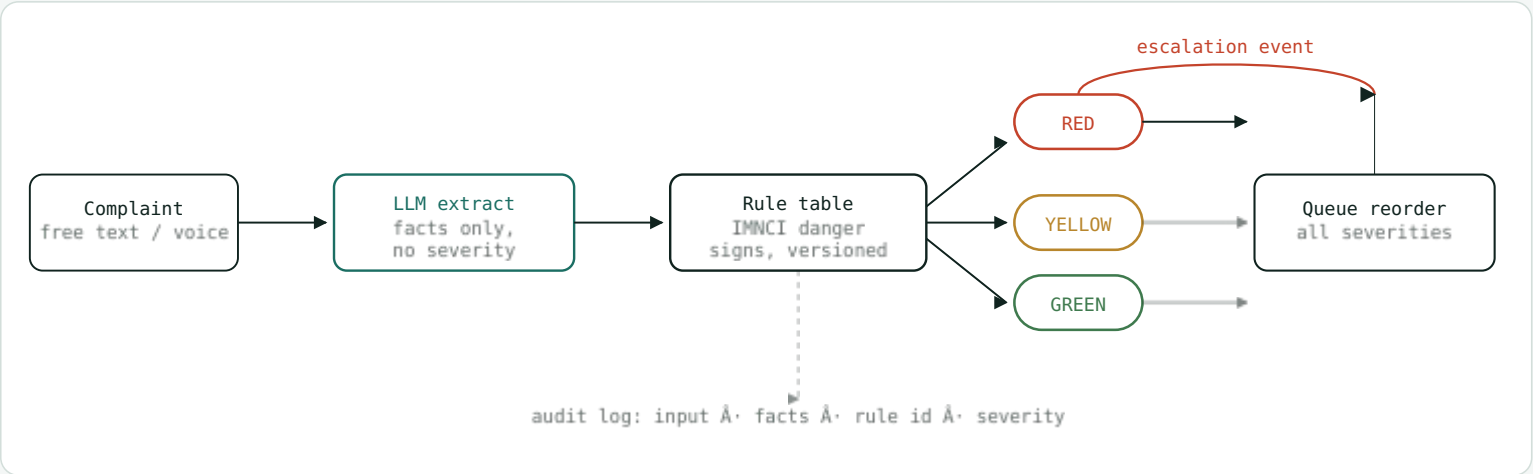
RESOURCE	HOLDS
Patient	Keyed by ABHA number “ the longitudinal thread across facility levels
Encounter	One per visit: who, where, when, facility level
Observation	Symptoms and vitals captured at triage

Relational tables

users , facilities , queue_tokens , triage_records , escalation_events , audit_log “ plus a single scheme_status enum (PMJAY / state / none) on Patient, the light-touch badge in place of a full scheme-verification engine.

05Triage logic “ the part judges will poke hardest

The LLM extracts facts. It never decides severity. A versioned rule table does that, and every decision is auditable “ that separation is what makes the triage defensible to a clinician.



Every RED severity does two things at once “ reorders the queue and fires an escalation to the facility doctor. Every decision, whatever the color, is written to an audit log naming which rule fired.

Offline degradation

If the AI service is unreachable, the LLM step is skipped entirely – the ASHA fills a structured symptom checklist instead, and the same rule table runs against it. The rule engine never depends on the model being available.

06 Offline-first ASHA app

Rural connectivity is intermittent by definition – an app that needs a live connection fails the problem statement on sight.

- PWA with a local write queue (IndexedDB): register, triage, and token creation all work with zero signal.
- The rule table + matcher run on-device – a plain JSON table, no server round trip required for severity.
- Sync on reconnect: server timestamps win, encounters are append-only, so conflicts are rare by construction.

07 Build order

Build the spine thin end-to-end first, then widen – never take one module to 100% while others sit at zero. If behind schedule, cut in reverse order: mocks first, offline second, the triage spine never.

- DAY 1 – SPINE
- Auth + roles, FHIR-shaped Patient record, create-patient flow on the ASHA app. Deploy immediately and keep it deployed.
- DAY 1 / 2 – TRIAGE
- LLM extract – rule engine – severity, with the matched rule shown in the UI. The centerpiece – give it the most polish.
- DAY 2 – QUEUE & ESCALATION
- Severity-based token reorder; RED auto-fires an escalation event and notification.
- DAY 2 – DASHBOARD
- Four tiles from real data: triaged-by-severity, queue length, teleconsults done, red cases escalated.
- DAY 2 / 3 – OFFLINE
- ASHA flow works fully offline and syncs on reconnect. High demo payoff for the effort.
- DAY 3 – MOCKS & POLISH
- Teleconsult store-and-forward, scheme badge, mock data screens, seeded demo patients, rehearsed script.

08 Top risks

	RISK	MITIGATION
1	Scope creep – mocks quietly become "real" builds	The \$01 table is law; cut in reverse order under time pressure
2	Live demo fails on venue wifi	Recorded fallback video + seeded demo data; never demo cold on conference wifi

3	A clinician judge challenges the triage	LLM-extracts-only + visible matched rule + audit log, shown live
4	ABDM sandbox access is slow to obtain	Never on the critical path – MockAbdmClient from day one
5	Teleconsult video becomes a rabbit hole	Capped at store-and-forward; video is not the headline feature

An offline-capable, ASHA-operated, ABDM/FHIR-linked record that follows a rural patient across every facility level, with explainable danger-sign triage that reorders the queue and escalates emergencies – strengthening the public system, not replacing it.

THE PITCH THIS ARCHITECTURE IS BUILT TO SUPPORT